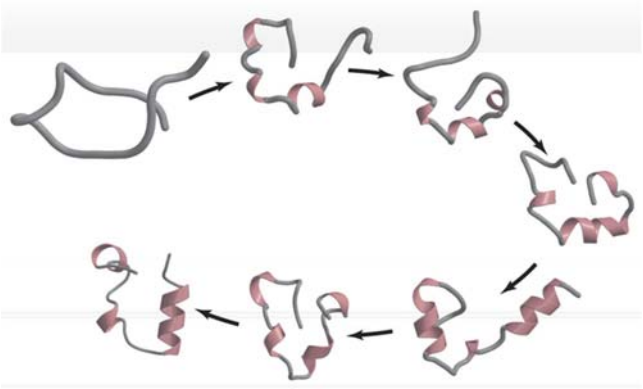
Maturation, transport and degradation of proteins

Protein folding



- During the translation process, the emerging polypeptide chain (N-ter) starts the folding process aimed to achieving the native conformation

- As we saw in previous lessons, the protein folding process is sequential and cooperative. In other words, as some regions acquire their secondary conformations, it is easier for other regions to do so as well.

- Certain Aa sequences fold into β-helices or sheets, drived by the restrictions that are determined by the angles of rotation in the polypeptide backbone.

- But not all proteins are capable of doing this path spontaneously

- Many require the assistance of other proteins, **called chaperones**, which interact with the partially folded or misfolded regions, facilitating the achievement of the final conformation

- **The two best-known families** of proteins that assist folding **are Hsp70 (heat shock proteins) and chaperonins**

- Hsp70 are proteins whose expression is induced during heat stress.

- Its function consists of interacting with proteins during their folding process, creating a microenvironment that favors the process. They can also join poorly folded regions, favoring their denaturation so that they refold correctly.

- They also intervene in protein transport processes to organelles, preventing them from folding until they have passed inside it

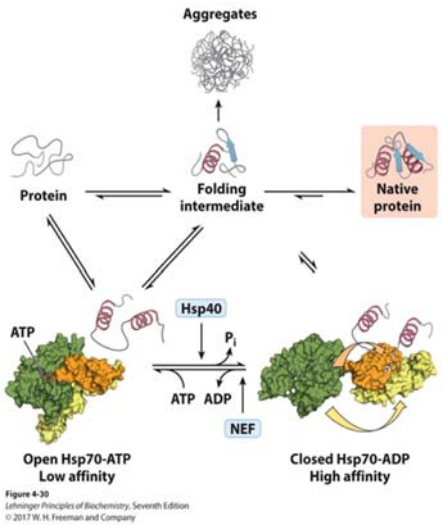


Figure 4-30
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- **Chaperonins** are actually multiprotein complexes, which are involved in the folding of proteins that do not fold spontaneously

- One of the best known chaperonins in prokaryotes is the GroEL / GroES complex, whose eukaryotic analog is the Hsp60 complex

- They are made up of several rings that form two chambers.

- The misfolded protein is trapped by the complex through hydrophobic interactions and passes into the chamber where it has time to fold before the ATP bound to the subunits that form the ring is hydrolyzed.

- This hydrolysis of ATP promotes a conformational change in GroES that allows the exit of the already folded protein

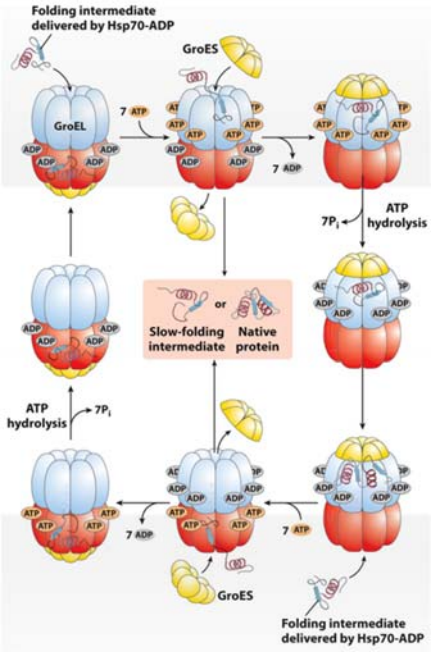


Figure 4-31a
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Targeting and protein degradation

- Proteins perform their functions in very different places in the cell: cytoplasm, lysosomes, membrane, nucleus, etc.

- Except for those synthesized in the mitochondria, most of the proteins are synthesized in membrane-associated ribosomes (RER), so they must be targeted to their final destination.

- The signals that define the final destination are located in the protein itself, in form of a short sequence of amino acids, called the signal sequence, which is frequently located in the amino-terminal region

- To demonstrate the importance of these sequences, chimeric proteins have been generated in which the amino-terminal region of a protein is replaced by the sequence of another protein whose destination must be a certain organelle (for example the lysosome). The result is that the chimeric protein appears in the lysosome.

- Similarly, the degradation of proteins, necessary for recycling, and as a mechanism for regulating activity, is a process whose selectivity is determined by signals contained in the amino acid sequence itself.

Targeting to the endoplasmic reticulum

- Most lysosomal, membrane or secretory proteins have an amino-terminal signal sequence that causes them to translocate into the RER.

- By analyzing hundreds of these sequences it has been possible to determine a common amino acid pattern (consensus sequence)

- At the C-ter end of the peptide there is an amino acid that constitutes the point of proteolysis that eliminates the peptide once the protein is located in the ER

Inner membrane proteins																								cleavage site												
Phage fd, major coat protein	Met	Lys	Lys	Ser	Leu	Val	Leu	Lys	Ala	Ser	Val	Ala	Val	Ala	Thr	Leu	Val	Pro	Met	Leu	Ser	Phe	Ala	Ala	Glu	--										
Phage fd, minor coat protein																																				
							Met	Lys	Lys	Leu	Leu	Phe	Ala	Ile	Pro	Leu	Val	Val	Pro	Phe	Tyr	Ser	His	Ser	Ala	Glu	--									
Periplasmic proteins																																				
Alkaline phosphatase				Met	Lys	Gln	Ser	Thr	Ile	Ala	Leu	Ala	Leu	Leu	Pro	Leu	Leu	Phe	Thr	Pro	Val	Thr	Lys	Ala	Arg	Thr	--									
Leucine-specific binding protein	Met	Lys	Ala	Asn	Ala	Lys	Thr	Ile	Ile	Ala	Gly	Met	Ile	Ala	Leu	Ala	Ile	Ser	His	Thr	Ala	Met	Ala	Asp	Asp	--									
β-Lactamase of pBR322	Met	Ser	Ile	Gln	His	Phe	Arg	Val	Ala	Leu	Ile	Pro	Phe	Phe	Ala	Ala	Phe	Cys	Leu	Pro	Val	Phe	Ala	His	Pro	--									
Outer membrane proteins																																				
Lipoprotein							Met	Lys	Ala	Thr	Lys	Leu	Val	Leu	Gly	Ala	Val	Ile	Leu	Gly	Ser	Thr	Leu	Leu	Ala	Gly	Cys	Ser	--						
LamB				Leu	Arg	Lys	Leu	Pro	Leu	Ala	Val	Ala	Val	Ala	Ala	Gly	Val	Met	Ser	Ala	Gln	Ala	Met	Ala	Val	Asp	--									
OmpA	Met	Met	Ile	Thr	Met	Lys	Lys	Thr	Ala	Ile	Ala	Ile	Ala	Val	Ala	Leu	Ala	Gly	Phe	Ala	Thr	Val	Ala	Gln	Ala	Ala	Pro	--								

- The length of the sequence ranges between 13-36 amino acids and the following characteristics are met

  - hydrophobic amino acids are abundant

  - a positively charged amino acid appears at the N-ter end

  - at the C-ter region there is a short sequence of polar amino acids with small side chains (Ala)

Translation mechanism at the ER

1) synthesis starts on free ribosomes

2) once the signal sequence (N-ter) has been synthesized, it emerges from the ribosome

3) the signal sequence binds to signal recognition particle ribonucleoprotein (SRP) and translation is stopped

4) the SRP binds GTP and the SRP-GTP complex drives the ribosome towards the SRP receptor located on the ER membrane. This allows the signal peptide to cross the membrane through the peptide translocation complex

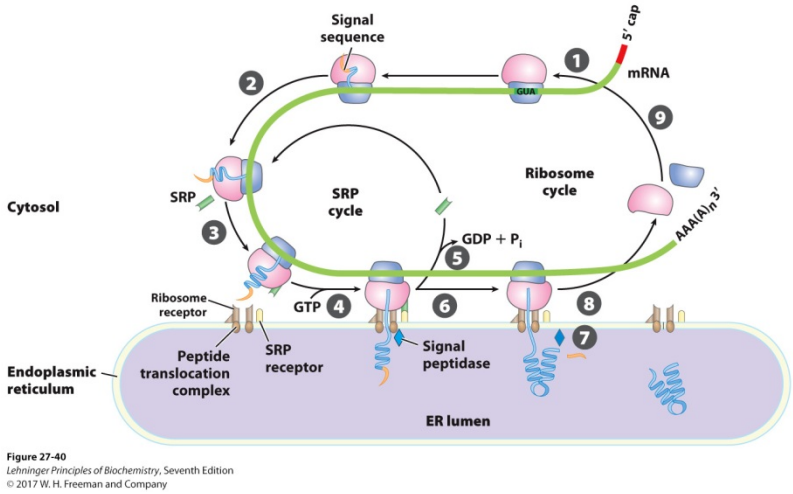


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5) GTP is hydrolyzed and SRP dissociates from the ribosome, which restarts translation

6) translation continues until complete protein synthesis is complete

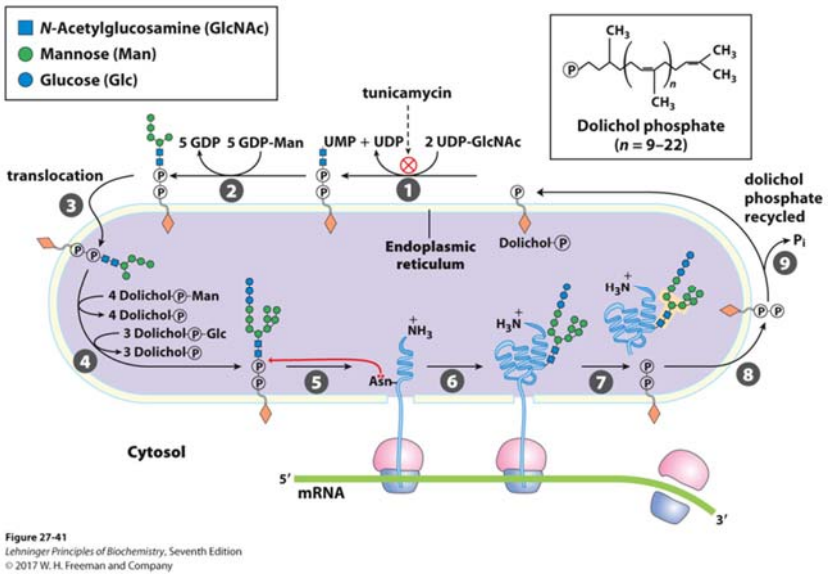
7) a peptidase proteolyzes the signal peptide

8) the ribosome dissociates from the ER membrane and is recycled (9)

Glycosilation

- During the translation at the ER the protein is chemically modified and starts to fold

- An important step of the chemical modifications consist of glycosidations, the addition of oligosaccharides that bind to the side chain of Asn



- Although the sequence of the oligosaccharide is different from protein to protein, there is a common core that is synthesized and later joined to the protein in the ER

- The glycosylation of the protein is an essential stage in its maturation towards the final conformation.

- When the process is disrupted, a stress is generated inside the reticulum and triggers a cellular response called UPR (unfolded protein response) aimed to reactivate the process or, if this is not achieved, to promote cell to enter in apoptosis

- Subsequently, the protein travels from the ER to the Golgi in vesicles

- In the Golgi, oligosaccharides are added through O-glucosides bonds

- Finally, through mechanisms that are not well known, proteins leave the Golgi to go to their final destinations.

Lysosomal targeting

- One of the best known transport is that of lysosome hydrolases

- These proteins, when they reach the Golgi, still contain a "tag" that causes them to be recognized by a phosphotransferase that phosphorylates a terminal mannose of the oligosaccharide

- The presence of one or more mannose 6-phosphate (M6P) residues is the structural signal that drives the protein to the lysosomes

- A receptor protein, located in the Golgi membranes, recognizes M6P located at the hydrolase. This interaction occurs at a pH range of 6-6.5.

- Then, the hydrolase-receptor complexes are coated by clathrin, which gives rise to the formation of transfer vesicles that are drived towards the lysosome.

- Inside the lysosome the pH is acidic (3-4) due to the continuous pumping of protons from the cytoplasm to the lumen

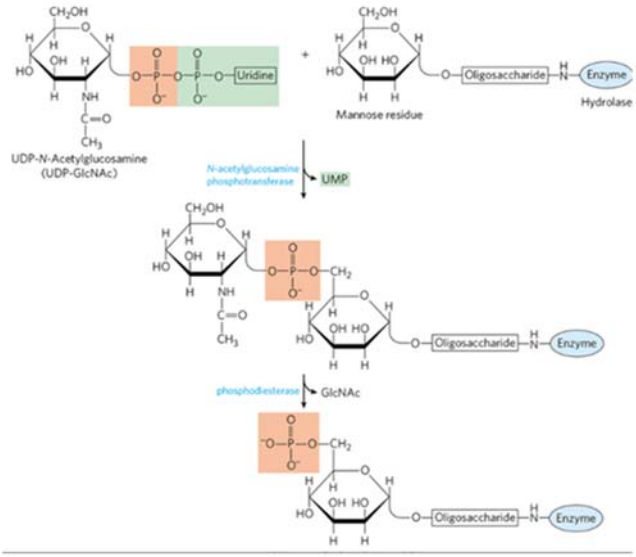
- This low pH causes the dissociation of the hydrolase from the M6P receptor.

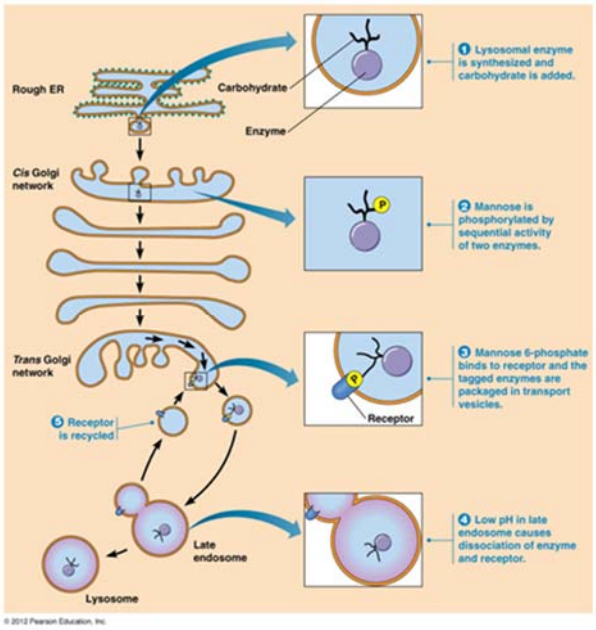
- Immediately, the M6P tag in the hydrolase is eliminated by a phosphatase and the receptor is also recycled towards the Golgi in the interior of vesicles. These two events prevent the hydrolase from being returned to the Golgi.

- This transport process is essential for the lysosome function to degrade protein and lipid. Impairments in this transport cause the malfunction of the lysosomes and the accumulation of waste products in their interior that, finally, will lead to cell dysfunction and serious diseases such as sphingolipidosis, glycogenosis or mucopolysaccharidosis

- The targeting of proteins towards the mitochondria is directed by specific sequences located in the N-ter region.

- In this case, the protein binds to chaperones that drive the protein to the mitochondria, where specific receptors are located, and promote its translocation to the interior.





Transport to the nucleus

- There are a large number of proteins (RNA and DNA polymerases, histones, topoisomerases, transcription factors) that are synthesized in cytoplasmic ribosomes and are then imported into the nucleus

- Nuclear proteins contain a signal sequence called NLS (nuclear localization sequence) that remains in the protein even after transport.

- The NLS consists of 4-8 amino acids among which the basic amino acids (R or K) are abundant

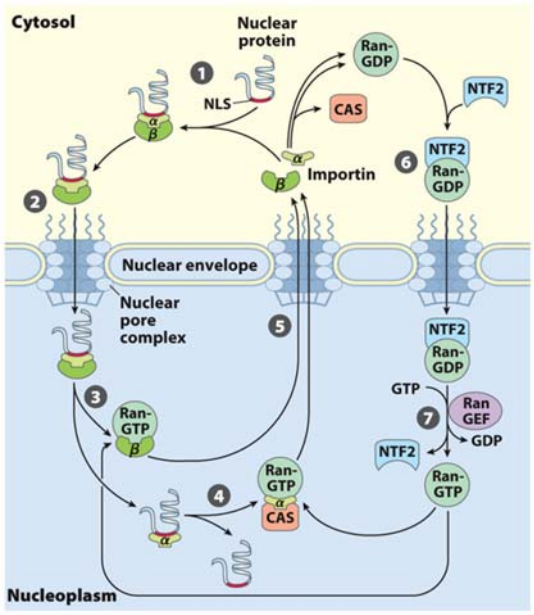


Figure 27-44a
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- Targeting of nuclear proteins is mediated by different proteins such as importins and the GTPase Ran (Ras related nuclear protein

- Importins alpha and beta bind to the protein in the cytoplasm and transport it to the nuclear pore complex where they translocate inside

- Once in the nucleus, the Ran protein binds to beta importin, separating it from the imported protein.

- The complex of nuclear protein and importin alpha binds to the Ran and Cas protein (cellular apoptosis susceptibility protein) causing the release of the nuclear protein

- Both importins, alpha and beta, are transferred to the cytoplasm where Ran hydrolyzes the GTP and releases the importins, being ready for a new importation

- Ran-GDP is recycled to the nucleus by nuclear transport factor 2 (NTF2)

- Within the nucleus an exchange factor (Ran-GEF) exchanges GDP for GTP.

Protein degradation

- Protein degradation is a critical process to maintain protein homeostasis

- Prevents the accumulation of unwanted aberrant proteins and allows amino acid recycling

- The half-life of eukaryotic proteins varies from about 30 seconds to several days. For example, hemoglobin has a half-life similar to that of the erythrocyte that contains it (about 110 days).

- Proteins that are rapidly degraded are those that contain damage or sequence errors.

- Proteins that have critical regulatory roles also have short half-life.

- Degradation systems in eukaryotes are of two types:

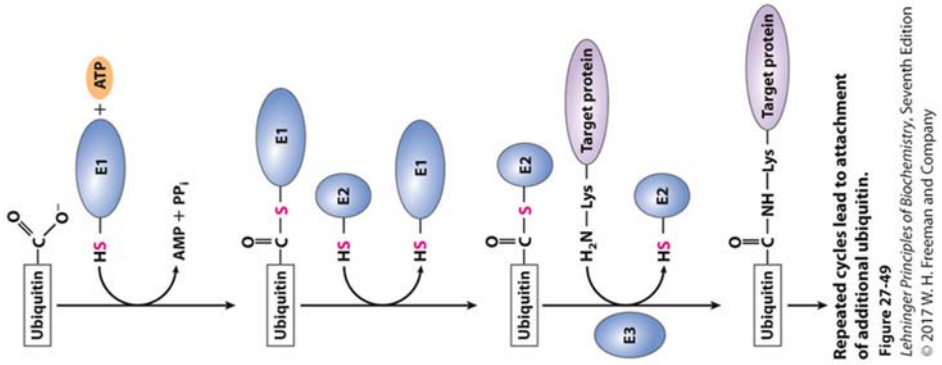
  - those that take place in lysosomes, which degrade membrane proteins, extracellular proteins and proteins with long half-lives

  - those that are dependent on ATP hydrolysis

- ATP dependent systems involve the protein called ubiquitin which, as its name suggests, is present in all living beings

- Ubiquitin has 76 amino acids and is identical in most living things.

- Ubiquitin binds covalently to lysine in proteins selected for destruction through an ATP-dependent pathway in which 3 enzymes are involved

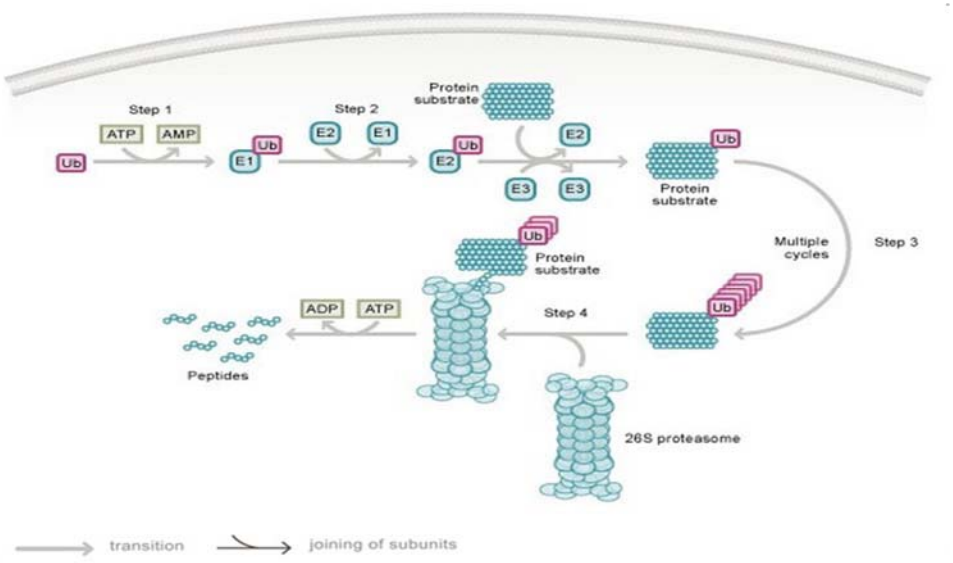


- The protein ubiquitination process can occur several times which generates polyubiquitin tails in the target protein

- Subsequently, the ubiquitinated proteins are degraded within a complex known as the proteasome

- The proteasome is made up of two copies of a complex that contains 32 different subunits

- Ubiquitinated proteins are proteolyzed inside the proteasome



- We still don't know exactly what are the signals that make a protein to be ubiquitinated and therefore degraded

- It seems that some amino acid sequences make the protein more prone to ubiquitination.

TABLE 27-9 Relationship between Protein Half-Life and Amino-Terminal Amino Acid Residue

Amino-terminal residue	Half-life[a]
Stabilizing	
Ala, Gly, Met, Ser, Thr, Val	> 20 h
Destabilizing	
Gln, Ile	~30 min
Glu, Tyr	~10 min
Pro	~7 min
Asp, Leu, Lys, Phe	~3 min
Arg	~2 min

Source: Information from A. Bachmair et al., *Science* 234:179, 1986.
[a]Half-lives were measured in yeast for the β-galactosidase protein modified so that in each experiment it had a different amino-terminal residue. Half-lives may vary for different proteins and in different organisms, but this general pattern appears to hold for all organisms.

- Impairments in protein ubiquitination and degradation are at the basis of many diseases
- Thus the alteration in the degradation of proteins involved in the cell cycle (products of oncogenes) give rise to the appearance of tumors
- Also, in most cases of neurodegenerative diseases (Alzheimer, Parkinson, Huntington ...) it can be observed that a common feature of all of them is the presence of aberrant protein aggregates whose accumulation is due, among other causes, to disability of the cell to get rid of them